

Tcpdump PCAP Investigation

PCAP file: tcpdump_sample01.pcap

Set the PCAP variable: Create a shell variable for the PCAP filename. This reduces typing errors and makes the workflow easier to repeat.

```
PCAP="tcpdump-sample01.pcap"
```

Count total packets

```
tcpdump -r "$PCAP" -nn 2>/dev/null | wc -l
```

Identify the main internal host that dominates the traffic.

```
tcpdump -tt -r "$PCAP" -n tcp | cut -d " " -f 3 | cut -d "." -f 1-4 | sort | uniq -c |  
sort -nr  
  
tcpdump -tt -r "$PCAP" -n tcp | cut -d " " -f 5 | cut -d "." -f 1-4 | sort | uniq -c |  
sort -nr
```

Count TCP, UDP, HTTP, HTTPS, and DNS traffic.

```
tcpdump -r "$PCAP" -nn 'tcp' 2>/dev/null | wc -l  
tcpdump -r "$PCAP" -nn 'udp' 2>/dev/null | wc -l  
tcpdump -r "$PCAP" -nn 'tcp port 80' 2>/dev/null | wc -l  
tcpdump -r "$PCAP" -nn 'tcp port 443' 2>/dev/null | wc -l  
tcpdump -r "$PCAP" -nn 'udp port 53' 2>/dev/null | wc -l
```

Extract HTTP requests: Search readable HTTP headers and methods.

```
tcpdump -r "$PCAP" -nn -A 'tcp port 80' 2>/dev/null \  
| grep -aE "GET |POST |Host:|User-Agent:|Referer:|Content-Type:|Content-Length:|Content-  
Key:"
```

Investigate the delivery server - Focus on the host that served the suspicious executable.

```
tcpdump -r "$PCAP" -nn -A 'host 103.232.55.148 and tcp port 80' 2>/dev/null \  
| grep -aE "GET |HTTP:|Host:|Content-Type:|Content-Length:|Last-  
Modified:|Server:|Location:"
```

Confirm Windows executable evidence: Search the payload for the Windows executable magic header.

```
tcpdump -r "$PCAP" -nn -X 'host 103.232.55.148 and tcp port 80' 2>/dev/null \  
| grep -a -A5 -B5 "MZ"
```

Focus on the exact malware download: Filter the communication between the victim and delivery server.

```
tcpdump -r "$PCAP" -nn -tttt -A \  
'host 103.232.55.148 and tcp port 80 and host 10.0.0.168' 2>/dev/null \  
| grep -aE "2021-|GET /service/.audiodg.exe|HTTP/1.1 200 OK|Content-Length:  
600576|Content-Type: application/x-msdownload"
```

Investigate post-download behavior: Search for HTTP POST requests after the download.

```
tcpdump -r "$PCAP" -nn -tttt -A 'tcp port 80' 2>/dev/null \  
| grep -aE "POST |Host:|User-Agent:|Content-Type:|Content-Encoding:|Content-Key:|Content-  
Length:"
```

Inspect the suspicious POST destination: Focus only on the suspected callback host.

```
tcpdump -r "$PCAP" -nn -tttt -A 'host 136.243.159.53 and tcp port 80' 2>/dev/null
```

Key HTTP Evidence

```
GET /service/ HTTP/1.1
Host: 103.232.55.148
User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64; Trident/7.0; rv:11.0) like Gecko

GET /service/.audiodg.exe HTTP/1.1
Host: 103.232.55.148

HTTP/1.1 200 OK
Content-Length: 600576
Content-Type: application/x-msdownload

POST /~element/page.php?id=484 HTTP/1.0
Host: 136.243.159.53
User-Agent: Mozilla/4.08 (Charon; Inferno)
Content-Type: application/octet-stream
Content-Encoding: binary
Content-Key: 4FE0A620
```

Indicators of Compromise

Type	Indicator
Victim IP	10.0.0.168
Victim hostname	GSNTPAWQ
Delivery server	103.232.55.148
Download URI	/service/.audiodg.exe
Downloaded filename	.audiodg.exe
Download content type	application/x-msdownload
Download size	600,576 bytes
Callback/C2 IP	136.243.159.53
Callback URI	/~element/page.php?id=484
Suspicious User-Agent	Mozilla/4.08 (Charon; Inferno)
Suspicious header	Content-Key: 4FE0A620
MD5 of extracted executable	da4f1a581f73ae0984cc58b61e12bd4c
SHA1 of extracted executable	4fff997f6479d3535b093abe1a5e297a3548d959
SHA256 of extracted executable	f485d1a65ccf9f857baa49725d337c15e8aa34515b85c8ef59a72afad7b85249

Note on hashes: The listed hashes apply only if the executable is correctly extracted from the HTTP response body. Students should not calculate a hash from the entire TCP stream or from HTTP headers plus payload.